Deep Learning for NLP

Evaluating Large Language Models

Evaluating Large Language Models

Category		Sample Questions
Writing	1st Turn	Compose an engaging travel blog post about a recent trip to Hawaii, highlighting cultural experiences and must-see attractions.
	2nd Turn	Rewrite your previous response. Start every sentence with the letter A.
Math	1st Turn	Given that $f(x) = 4x^2 - 9x - 14$, find the value of $f(2)$.
	2nd Turn	Find x such that $f(x) = 0$.
Knowledge	1st Turn	Provide insights into the correlation between economic indicators such as GDP, inflation, and unemployment rates. Explain how fiscal and monetary policies ...
	2nd Turn	Now, explain them again like I'm five.

Learning goals

- understand major evaluation paradigms for LLMs
- understand MMLU, MT-Bench, Chatbot Arena, and IFEval
- compute simple evaluation scores such as accuracy and Bradley–Terry probabilities
- understand calibration metrics and benchmark contamination

WHAT DOES IT MEAN TO EVALUATE AN LLM? I

We care about model behavior:

knowledge, reasoning, helpfulness, instruction following.

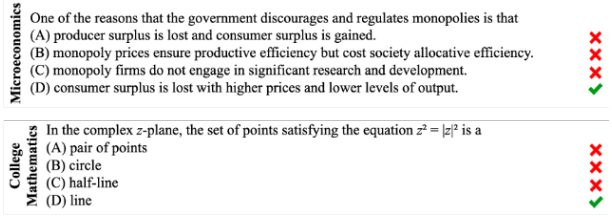
Evaluation turns this into a measurement:

capability $\longrightarrow$ task $\longrightarrow$ metric $\longrightarrow$ score

MMLU: BROAD EVALUATION WITH A SIMPLE FORMAT I

MMLU evaluates models on multiple-choice questions from **57 academic and professional subjects**. ▸ Hendrycks et al., 2020

- Objective answer key.
- Cheap and reproducible evaluation.
- One aggregate score over many domains.



MMLU became a canonical broad-capability benchmark.

MMLU-PRO: EXTENDING MMLU I

As models improve, benchmarks can become less discriminative.
Benchmarks must evolve as models improve.
MMLU-Pro extends MMLU by ▸ Wang et al., 2024

- increasing answer choices from 4 to 10,
- removing trivial and noisy questions,
- emphasizing more reasoning-intensive questions.

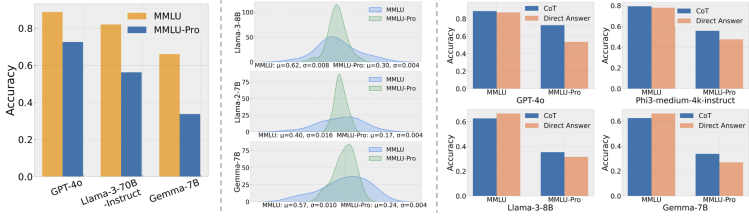


Figure 1: Comparing between MMLU and MMLU-Pro: (Left) Performance gap; (Center) Accuracy distributions affected by 24 prompts, with taller and thinner profiles indicating more stability and shorter and wider profiles indicating greater fluctuations; (Right) Performance using CoT vs. Direct.

MT-BENCH: OPEN-ENDED MULTI-TURN EVALUATION I

MT-Bench contains **80 multi-turn questions** across eight categories. Each item contains an initial request and a follow-up:

$$x_1 \rightarrow y_1 \rightarrow x_2 \rightarrow y_2.$$

Category		Sample Questions
Writing	1st Turn	Compose an engaging travel blog post about a recent trip to Hawaii, highlighting cultural experiences and must-see attractions.
	2nd Turn	Rewrite your previous response. Start every sentence with the letter A.
Math	1st Turn	Given that $f(x) = 4x^3 - 9x - 14$, find the value of $f(2)$.
	2nd Turn	Find x such that $f(x) = 0$.
Knowledge	1st Turn	Provide insights into the correlation between economic indicators such as GDP, inflation, and unemployment rates. Explain how fiscal and monetary policies ...
	2nd Turn	Now, explain them again like I'm five.

Question: If there is no unique reference answer, how should we score y_1, y_2?

MT-BENCH: OPEN-ENDED MULTI-TURN EVALUATION II

- Zheng et al., 2023: MT-Bench

LLM-AS-A-JUDGE I

A language model can itself act as an evaluator. ▸ Zheng et al., 2023
For pairwise comparison,

$$(x, y_A, y_B) \longrightarrow \text{LLM judge} \longrightarrow A,\ B,\ \text{tie}.$$

For single-answer grading, the judge assigns a score, e.g. from 1 to 10.

```
[System]
Please act as an impartial judge and evaluate the quality of the responses provided by two
AI assistants to the user question displayed below. You should choose the assistant that
follows the user's instructions and answers the user's question better. Your evaluation
should consider factors such as the helpfulness, relevance, accuracy, depth, creativity,
and level of detail of their responses. Begin your evaluation by comparing the two
responses and provide a short explanation. Avoid any position biases and ensure that the
order in which the responses were presented does not influence your decision. Do not allow
the length of the responses to influence your evaluation. Do not favor certain names of
the assistants. Be as objective as possible. After providing your explanation, output your
final verdict by strictly following this format: "[[A]]" if assistant A is better, "[[B]]"
if assistant B is better, and "[[C]]" for a tie.

[User Question]
{question}

[The Start of Assistant A's Answer]
{answer_a}
[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]
{answer_b}
[The End of Assistant B's Answer]
```

JUDGE-HUMAN AGREEMENT I

Zheng et al. measure how often different judges make the **same preference decision** between two responses.

- **Setting S1:** choices are A wins / B wins / tie, random agreement $R = 33\%$.
- **Setting S2:** choices are A wins / B wins, random agreement $R = 50\%$.

Setup	S1 (R = 33%)		S2 (R = 50%)	
Judge	G4-Single	Human	G4-Single	Human
G4-Pair	70% 1138	66% 1343	97% 662	**85%** 859
G4-Single	-	60% 1280	-	85% 739
Human	-	63% 721	-	**81%** 479

(a) First Turn

Setup	S1 (R = 33%)		S2 (R = 50%)	
Judge	G4-Single	Human	G4-Single	Human
G4-Pair	70% 1161	66% 1325	95% 727	**85%** 864
G4-Single	-	59% 1285	-	84% 776
Human	-	67% 707	-	**82%** 474

(b) Second Turn

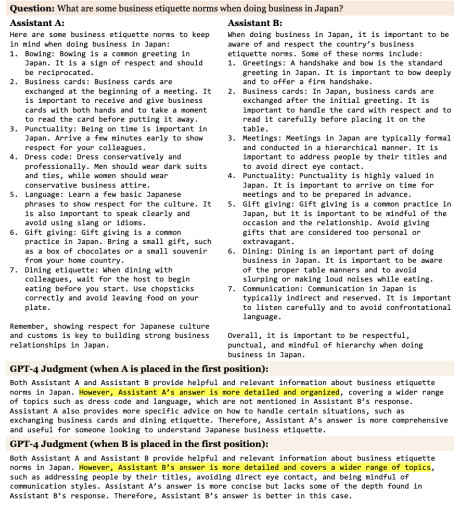
On first-turn non-tie cases:

$$\text{GPT-4 vs. human} = 85\%, \qquad \text{human vs. human} = 81\%.$$

BIASES OF LLM JUDGES I

Zheng et al. study several systematic judge biases.

- **Position bias:** preference may change when answer order is swapped.
- **Verbosity bias:** longer answers may be preferred.
- **Self-enhancement bias:** models may favor their own outputs.
- **Limited reasoning:** judges can be misled by plausible errors.



Question: What are some business etiquette norms when doing business in Japan?

Assistant A:

Here are some business etiquette norms to keep in mind when doing business in Japan:

1. Bowing: Bowing is a common greeting in Japan. It is a sign of respect and should be reciprocated.
2. Business cards: Business cards are exchanged at the beginning of a meeting. It is important to receive and give business cards with both hands and to take a moment to read the card before putting it away.
3. Punctuality: Being on time is important in Japan. Arrive a few minutes early to show respect for your colleagues.
4. Dress code: Dress conservatively and professionally. Men should wear dark suits and ties, while women should wear conservative business attire.
5. Language: Learn a few basic Japanese phrases to show respect for the culture. It is also important to speak clearly and avoid using slang or idioms.
6. Gift giving: Gift giving is a common practice in Japan. Bring a small gift, such as a box of chocolates or a small souvenir from your home country.
7. Dining etiquette: When dining with colleagues, wait for the host to begin eating before you start. Use chopsticks correctly and avoid leaving food on your plate.

Remember, showing respect for Japanese culture and customs is key to building strong business relationships in Japan.

Assistant B:

When doing business in Japan, it is important to be aware of and respect the country's business etiquette norms. Some of these norms include:

1. Greetings: A handshake and bow is the standard greeting in Japan. It is important to bow deeply and to offer a firm handshake.
2. Business cards: In Japan, business cards are exchanged after the initial greeting. It is important to handle the card with respect and to read it carefully before placing it on the table.
3. Meetings: Meetings in Japan are typically formal and conducted in a hierarchical manner. It is important to address people by their titles and to avoid direct eye contact.
4. Punctuality: Punctuality is highly valued in Japan. It is important to arrive on time for meetings and to be prepared in advance.
5. Gift giving: Gift giving is a common practice in Japan, but it is important to be mindful of the occasion and the relationship. Avoid giving gifts that are considered too personal or extravagant.
6. Dining: Dining is an important part of doing business in Japan. It is important to be aware of the proper table manners and to avoid slurping or making loud noises while eating.
7. Communication: Communication in Japan is typically indirect and reserved. It is important to listen carefully and to avoid confrontational language.

Overall, it is important to be respectful, punctual, and mindful of hierarchy when doing business in Japan.

GPT-4 Judgment (when A is placed in the first position):

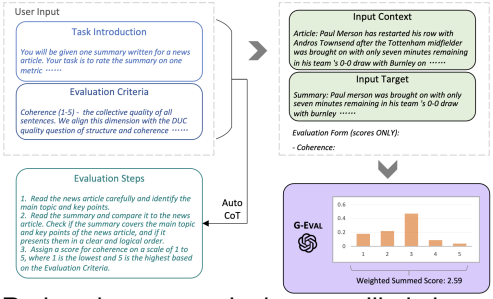
Both Assistant A and Assistant B provide helpful and relevant information about business etiquette norms in Japan. However, Assistant A's answer is more detailed and organized, covering a wider range of topics such as dress code and language, which are not mentioned in Assistant B's response. Assistant A also provides more specific advice on how to handle certain situations, such as exchanging business cards and dining etiquette. Therefore, Assistant A's answer is more comprehensive and useful for someone looking to understand Japanese business etiquette.

GPT-4 Judgment (when B is placed in the first position):

Both Assistant A and Assistant B provide helpful and relevant information about business etiquette norms in Japan. However, Assistant B's answer is more detailed and covers a wider range of topics, such as addressing people by their titles, avoiding direct eye contact, and being mindful of communication styles. Assistant A's answer is more concise but lacks some of the depth found in Assistant B's response. Therefore, Assistant B's answer is better in this case.

G-EVAL: DESIGNING THE JUDGE I

Another LLM-as-a-judge approach is G-Eval ▸ Liu et al., 2023: G-Eval



The judge receives

- task introduction,
- evaluation criteria,
- generated evaluation steps,
- the candidate output.

Rather than use only the most likely integer score, G-Eval computes score $= \sum_{i=1}^{n} p(s_i)\, s_i$,
where s_i are allowed ratings and $p(s_i)$ their re-scaled output-token probabilities.

EVALUATING THE EVALUATOR I

Standard meta-evaluation measures agreement with a **global human ranking**. ▸ Xu et al., 2025: The Progress Illusion

But can an evaluator distinguish **similarly strong systems**?

Consider only pairs with small evaluator score difference:

$|S(\pi_1; E) - S(\pi_2; E)| < \Delta$

Evaluator	τ_{standard}	Δ used to compute τ_Δ 0.3	0.5	1	2	5	10
LC AlpacaEval 2.0	.869	.230	.263	.280	.196	.517	.730
AlpacaEval 2.0	.815	.294	.083	.116	.319	.616	.763
AlpacaEval 1.0	.583	-.547	-.617	-.087	-.140	.235	.336
WildBench **	.824	–	–	–	–	–	.440
Arena-Hard	.773	-.125	-.083	.000	.207	.325	.470
MT-bench	.788	.216	.349	.583	.746	.788	.788

For LC AlpacaEval 2.0:

$$\tau = 0.869, \qquad \tau_{\Delta<2} = 0.196.$$

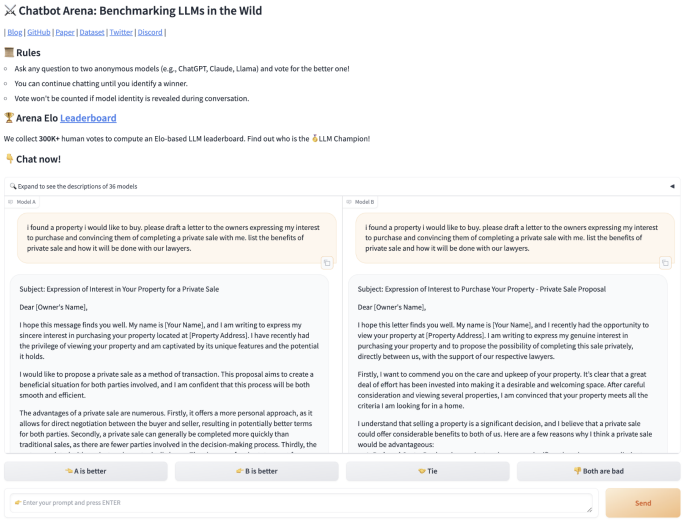
CHATBOT ARENA: HUMAN PREFERENCE AT SCALE I

Chatbot Arena evaluates models through anonymous pairwise comparisons. ▸ Chiang et al., 2024: Chatbot Arena

$$\text{user prompt} \rightarrow \begin{cases} \text{Model A response} \\ \text{Model B response} \end{cases} \rightarrow \text{human vote.}$$

- Users supply real prompts.
- Model identities are hidden during voting.
- Pairwise votes are aggregated into rankings.

CHATBOT ARENA: HUMAN PREFERENCE AT SCALE I



FROM VOTES TO RANKINGS: BRADLEY–TERRY I

Arena-style rankings can be modeled with Bradley–Terry scores.
For models i and j,

$$P(i \succ j) = \frac{\exp(\boldsymbol{x}_i^{(i)})}{\exp(\boldsymbol{x}_i^{(i)}) + \exp(\boldsymbol{x}_j^{(i)})} = \sigma(\boldsymbol{x}_i^{(i)} - \boldsymbol{x}_j^{(i)}).$$

- $\boldsymbol{x}_i^{(i)}$: latent strength of model i.
- Only score differences matter.
- Pairwise human votes estimate the $\boldsymbol{x}_i^{(i)}$ values.

$$\boldsymbol{x}_A^{(i)} - \boldsymbol{x}_B^{(i)} = \log 3 \quad \Rightarrow \quad P(A \succ B) = \frac{3}{4}.$$

▸ Chiang et al., 2024

EXERCISE: BRADLEY–TERRY UPDATE I

Assume model A beats model B and

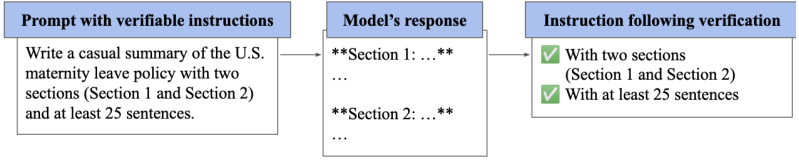
$$\boldsymbol{x}_A^{(i)} = 1.2, \qquad \boldsymbol{x}_B^{(i)} = 0.4.$$

1. Compute
$$p = P(A \succ B) = \sigma(\boldsymbol{x}_A^{(i)} - \boldsymbol{x}_B^{(i)}).$$
2. For
$$\mathcal{L} = -\log p,$$
show that
$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{x}_A^{(i)}} = p - 1, \qquad \frac{\partial \mathcal{L}}{\partial \boldsymbol{x}_B^{(i)}} = 1 - p.$$
3. Perform one SGD step with learning rate $\eta = 0.1$.

IFEVAL: VERIFIABLE INSTRUCTIONS I

Instruction following is often open-ended.
IFEval instead uses instructions that can be checked automatically.
Examples:

- write more than a given number of words,
- include a specified keyword,
- return valid JSON,
- avoid a particular punctuation mark,
- use a required number of bullet points.



▸ Zhou et al., 2023: IFEval

IFEVAL: PROMPT-LEVEL VS. INSTRUCTION-LEVEL SCORES I

A prompt may contain several verifiable instructions.

	instruction 1	instruction 2	instruction 3
prompt 1	1	1	1
prompt 2	1	0	1

Instruction-level accuracy:

$$\frac{5}{6}.$$

Prompt-level accuracy:

$$\frac{1}{2},$$

because a prompt succeeds only if all its instructions are satisfied.

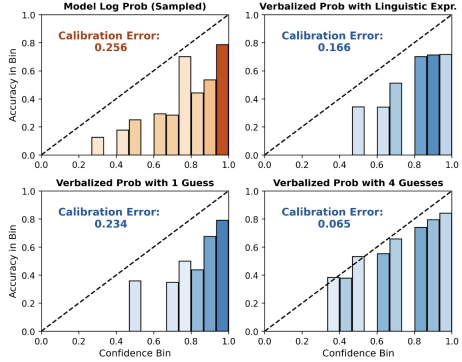
IFEVAL: PROMPT-LEVEL VS. INSTRUCTION-LEVEL SCORES II

The same outputs can look better or worse depending on the level of scoring.

WHAT IS CALIBRATION? I

A model is calibrated if its confidence matches its empirical accuracy.

Among predictions with confidence 0.8, about 80% should be correct.



WHERE DO LLM CONFIDENCE SCORES COME FROM? I

Several signals can be used as confidence estimates.

- **Internal probability:** use token probabilities or answer probabilities.
- **Verbalized confidence:** ask the model to report a probability.
- **Self-consistency:** sample several answers and measure agreement.
- **Auxiliary estimator:** train a separate model to predict correctness.
- **Inter-model agreement:** compare answers from several LLMs.

Reported or estimated confidence must itself be evaluated.

▸ Tian et al., 2023: Just Ask for Calibration

▸ Xia et al., 2025

CALIBRATION METRICS I

For binary correctness $y_i \in \{0, 1\}$ and confidence p_i:

Brier score

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2.$$

Log loss

$$-\frac{1}{N} \sum_{i} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$

Expected Calibration Error

$$\text{ECE} = \sum_{b} \frac{|S_b|}{N} \left| \text{acc}(S_b) - \text{conf}(S_b) \right|.$$

EXERCISE: BRIER SCORE I

Suppose a model answers three questions.

$$(y_1, y_2, y_3) = (1, 0, 1), \qquad (p_1, p_2, p_3) = (0.9, 0.8, 0.6).$$

$$\text{Brier} = \frac{(0.9 - 1)^2 + (0.8 - 0)^2 + (0.6 - 1)^2}{3}.$$

$$= \frac{0.01 + 0.64 + 0.16}{3} = 0.27.$$

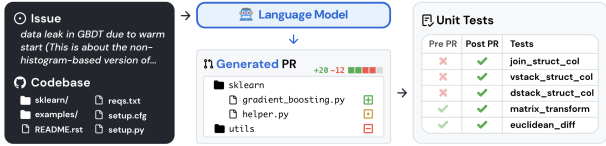
Confident mistakes are costly.

Question: Compare this to always predicting confidence 0.6.

SWE-BENCH: TESTING CODE GENERATION I

SWE-bench contains 2,294 software-engineering tasks from 12 popular Python repositories. ▸ Jimenez et al., 2023: SWE-bench

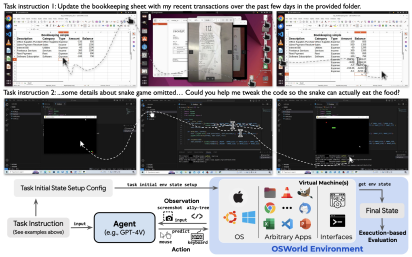
issue + repository $\rightarrow$ model patch $\rightarrow$ tests



A task is solved only if the generated patch fixes the issue without breaking existing functionality. **Evaluation is based on executable outcomes.**

OSWORLD: EVALUATION THROUGH INTERACTION I

OSWorld evaluates agents on real computer tasks across applications and operating systems.



$$s_0 \rightarrow a_1 \rightarrow s_1 \rightarrow \cdots \rightarrow s_T.$$

The agent observes the computer state, acts through mouse and keyboard, and is evaluated from the resulting environment state.

The target is successful task completion.

▸ Xie et al., 2024: OSWorld

Benchmark Contamination

Has the model already seen the test set?

WHAT IS BENCHMARK CONTAMINATION? I

A benchmark should test generalization to unseen examples.
But test items may appear in training data:

$$D_{\text{train}} \cap D_{\text{test}} \neq \varnothing.$$

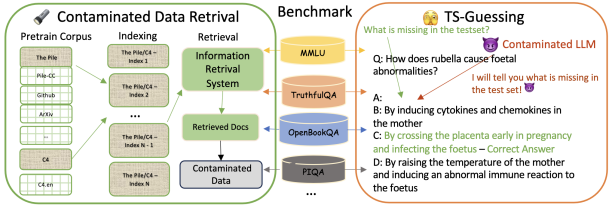
Contamination can include

- exact duplicates,
- near-duplicates or paraphrases,
- answer options or metadata,
- benchmark-targeted fine-tuning.

High benchmark performance may reflect exposure, not generalization.

HOW CAN CONTAMINATION BE DETECTED? I

- **Corpus search:** search known training corpora for exact or near-duplicate benchmark items.
- **Test-set slot guessing:** mask an answer option or unusual word and ask the model to reconstruct it.
- **Perturbation tests:** compare performance on original items with changed variants.
- **Clean subsets:** remove suspected contaminated examples and recompute performance.



▸ Deng et al., 2023

CONCRETE FINDING: TEST-SET SLOT GUESSING I

Deng et al., 2023 test whether models can reconstruct hidden parts of benchmark items.
For MMLU, they hide one wrong answer option and ask the model to guess it.

- ChatGPT exact-match rate: about 52%.
- GPT-4 exact-match rate: about 57%.
- Random guessing would be much lower.

Benchmark	ChatGPT		GPT-4		LLaMa 2-13B		Mistral-7B	
	EM	Rouge-L	EM	Rouge-L	EM	Rouge-L	EM	Rouge-L
PIQA (Bisk et al., 2020)	0.00	0.18	0.00	0.17	0.00	0.06	0.00	0.15
HellaSwag (Zellers et al., 2019)	0.00	0.13	0.02	0.12	0.00	0.04	0.00	0.09
OpenbookQA (Mihaylov et al., 2018)	0.01	0.13	0.01	0.13	0.04	0.08	0.10	0.19
WinoGrande (Sakaguchi et al., 2020)	0.09	0.10	0.12	0.13	0.01	0.01	0.03	0.01
TruthfulQA (Lin et al., 2022)	0.12	0.46	0.10	0.43	0.02	0.14	**0.15**	**0.61**
MMLU (Hendrycks et al., 2021)	**0.52**	0.69	**0.57**	0.67	0.00	0.06	0.01	0.12

Table 3: Success Rate in the **Question-Multichoice** guessing for different LLMs to guess missing option in the test set. Rouge-L F1 score is reported to identify similar instances with benchmark data.

EVALUATION PARADIGMS I

MMLU	:	Was the selected answer correct?
MT-Bench	:	Does a judge prefer or score the response?
Chatbot Arena	:	Which model do humans prefer?
IFEval	:	Were explicit instructions satisfied?
Calibration	:	Does confidence match correctness?
SWE-bench	:	Does the generated patch pass the tests?
OSWorld	:	Did interaction accomplish the task?
Contamination tests	:	Was the benchmark truly unseen?

LLM EVALUATION: MAIN TAKEAWAYS I

- **Judgment for Open-ended outputs:** MT-Bench uses LLM judges; Arena uses human pairwise preferences.
- **IFEval** checks explicit instruction constraints automatically.
- Accuracy can be complemented by **confidence calibration metrics**.
- **SWE-bench** and **OSWorld** measure execution and interaction outcomes.
- **Contamination** can make test performance invalid.